METHODS & EVIDENCE

The logic behind the conversation.

Risk Continuum CKM

Clinical logic, Level rules, RSS weights, simulator methods, engineering and dated evaluation evidence.

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Released-engine reference: 18baf7a8651b10cb4f047072590a5c8699e293c5
RSS methodology: rss-methodology-v2
Application PRs #79 and #80 are separately identified proposals, not released behavior.

This appendix includes the full technical sections available on the methods page, including the ordered Level rules and every RSS band. It describes implementation and software verification. Independent prospective clinical effectiveness and external prognostic validation of RCCKM-specific frameworks have not been established.

riskcontinuum.health/methods.html

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One assessment, distinct kinds of information.

RCCKM combines published risk equations and clinical frameworks with a prespecified interpretation layer. Each output answers a different question. The platform's own Levels and Risk Signal Score (RSS) organize findings for review; their clinical usefulness and prognostic properties require evaluation in their own right.

Meaning and authority of each output

Output	Question it addresses	Basis
PREVENT	What is the modeled probability of a cardiovascular event over a defined horizon?	Published AHA equations, with model eligibility and input-domain checks. [1]
AHA CKM stage	Where do cardiovascular, kidney and metabolic findings sit in the CKM framework?	AHA's disease and risk staging framework, implemented separately from RCCKM Level. [2]
KDIGO G/A context	What do kidney function and albuminuria show, and is chronic kidney disease established?	KDIGO classification plus explicit chronicity and certainty handling. [3]
RCCKM Level	Which prespecified clinical pattern best describes the reviewed findings?	RCCKM's ordered rule hierarchy, summarized below. A Level does not carry an event probability.
Risk Signal Score	Which selected findings contribute to the visible burden summary?	RCCKM's fixed ordinal points and overlap rules. The total has no validated 100-point denominator.

How source evidence becomes a platform rule

Published sources inform risk equations, disease definitions, measured findings and treatment pathways. RCCKM then specifies how to reconcile those inputs, prioritize pathways and present them together. The choice to group a pattern into a particular Level, assign an RSS weight, or combine trial effects with PREVENT is a platform design decision. A citation to a guideline or trial supports its underlying evidence; it does not establish external validation of RCCKM's synthesis.

PREVENT probabilities, CKM stages, KDIGO categories, Level and RSS remain separate in the assessment. Domain-specific recommendations use the clinical findings and applicable pathways. RSS does not independently establish a diagnosis, treatment indication or target.

Proposed presentation update: probability and clinical findings

Under review, not a released change. Application draft PR #79 leads the main Next.js assessment with PREVENT probability and category, followed by separate disease/plaque findings, prioritized risk drivers and a visible input-availability cue. The legacy Level code and framework markers remain in technical details; existing exported notes, visual reports and patient handouts retain their current codes.

The proposed display does not place plaque and driver burden on a new shared severity scale. A person with CAC 5 and a person with CAC 0, severe hypercholesterolemia, diabetes and reduced eGFR can have different disease evidence, drivers and modeled probability. Neither person's position on one numbered line establishes the other's expected event risk. The legacy rule sequence below documents implementation behavior, not a calibrated ranking of those cases.

The data cue distinguishes supplied values, results needing review and values not supplied. Measured CAC 0 remains distinct from unmeasured CAC. Completed bounded or nonnumeric UACR screening remains visible without an invented exact ratio. Availability does not establish recency, diagnostic confirmation or an indication to obtain every missing test.

How the legacy rule pathway is assigned.

The engine first checks clinical ASCVD, established heart failure and quantitative coronary calcium. It then evaluates severe lipid states, kidney findings, long-term PREVENT risk and specified combinations of actionable or early signals. The first matching base rule determines the Level. A final modifier distinguishes Level 3C from 3B.

Level names and principal assignment patterns

Level	Principal pattern in the reviewed implementation
1No signals found	No earlier qualifying rule or counted mild signal matches, and the classifier's blocking PREVENT inputs are not missing. Read this in the context of the data available.
2AOne early signal	One counted mild signal, or an isolated kidney finding requiring diagnostic confirmation after higher-priority rules have been considered.
2BSeveral early signals	At least two counted mild signals without an earlier actionable, disease or plaque pathway.
3AHigh long-term risk	Qualifying long-term risk or a single-driver pathway: for example, 30-year PREVENT ASCVD ≥10% at ages 30-59 without qualifying additional burden, LDL-C 160-189 mg/dL, ApoB ≥120 mg/dL, or isolated current smoking. A specific inflammatory-context rule also maps here.
3BActionable drivers	A qualifying severe-lipid or clinical pathway, or a specified combination of atherogenic, metabolic, kidney, enhancer and/or PREVENT findings, before the 3C modifier.

Level names and principal assignment patterns

Level	Principal pattern in the reviewed implementation
3CCompounding drivers	A base 3B classification plus any of: 10-year PREVENT ASCVD ≥5%, established diabetes, unconfirmed diabetes-range A1c, or a qualifying kidney finding.
4Plaque present	Quantitative CAC >0 and <300, after clinical disease exclusions.
5Advanced plaque or disease	Clinical ASCVD, established heart failure, or quantitative CAC ≥300. The underlying disease context remains explicit; heart failure alone does not imply an ASCVD diagnosis or ASCVD secondary-prevention lipid targets.
Incomplete	No earlier qualifying rule matches, PREVENT ASCVD is unavailable, and at least one of systolic BP, total cholesterol, HDL-C, eGFR or BMI is recorded as a blocking missing input.

Pattern labels summarize the rule that fired. Level 3A can arise from a single-driver rule even when a 30-year estimate is unavailable or below 10%. The rule identifier and explanation supply the specific reason. Levels are prespecified categories; their ordering and boundaries have not been externally calibrated as a prognostic scale.

Information that keeps its meaning

Unmeasured CAC remains unmeasured. In this revision, numeric CAC drives the plaque-based Level 4/5 rules; qualitative incidental coronary calcium is retained in plaque context, RSS and applicable domain actions. CAC 0 does not remove a severe-hypercholesterolemia or suspected FH pathway. An isolated kidney abnormality can influence risk interpretation while CKD chronicity remains unestablished.

Technical appendix A · Complete base-rule sequence and signal definitions

Rules below follow the classifier's evaluation order. Each row is evaluated only if previous rows have not returned a classification. The final 3C modifier is applied only to a base 3B result. Signal counts are prespecified rule counters; correlated findings can appear in more than one counter. They are separate from RSS's overlap caps.

Ordered base rules, engine reference 18baf7a

Order	Condition	Base output and rule identifier
1	Clinical ASCVD.	5 LEVEL_5_CLINICAL_ASCVD

Ordered base rules, engine reference 18baf7a

Order	Condition	Base output and rule identifier
2	Established heart failure, HFpEF or HFrEF.	5 LEVEL_5_HEART_FAILURE
3	Quantitative CAC ≥300.	5 LEVEL_5_CAC_GE_300
4	Quantitative CAC >0.	4 LEVEL_4_CAC_POSITIVE
5	Canonical severe hypercholesterolemia (valid LDL-C ≥190 mg/dL or ApoB ≥140 mg/dL), or suspected FH/HeFH.	3B LEVEL_3B_SEVERE_ HYPERCHOLESTEROLEMIA_FH
6	Current kidney category G4/G5 plus A3; compatibility path: eGFR <30 and UACR ≥300.	3B LEVEL_3B_ADVANCED_KIDNEY_ ALBUMINURIA
7	Albuminuria plus at least two actionable drivers or at least one mild signal.	3B LEVEL_3B_ALBUMINURIA_PLUS_ BURDEN
8a	Age 30-59 and 30-year PREVENT ASCVD ≥10%, with HIGH 10-year PREVENT ASCVD category.	3B LEVEL_3B_HIGH_10Y_WITH_ ELEVATED_30Y
8b	The same elevated 30-year condition with any actionable driver or at least two mild signals. ApoB ≥120 plus premature family history receives a specific reason identifier.	3B LEVEL_3B_30Y_PREVENT_APOB_ FAMILY_HISTORY LEVEL_3B_30Y_PREVENT_PLUS_ BIOLOGIC_BURDEN
8c	The same elevated 30-year condition without the preceding additional burden.	3A LEVEL_3A_30Y_PREVENT
9	Age 40-59, CAC unmeasured, low short-term PREVENT (<3% or LOW category), ApoB ≥120 or LDL-C 150-<190, and at least two hidden-risk enhancers defined below.	3B LEVEL_3B_HIDDEN_RISK_CLUSTER
10	Age 40-59, CAC unmeasured, low short-term PREVENT, ApoB ≥100, premature family history, and at least two of: early glycemic risk, TG ≥150, valid LDL-C ≥130, or CKM context.	3B LEVEL_3B_APOB_FAMILY_ METABOLIC

Ordered base rules, engine reference 18baf7a

Order	Condition	Base output and rule identifier
11	Valid LDL-C 160-189.999 mg/dL. At least two actionable drivers or two mild signals gives 3B; otherwise 3A. The young family/metabolic helper changes the explanation within the 3B branch.	3B or 3A LEVEL_3B_LDL_160_189_CONVERGENCE LEVEL_3A_LDL_160_189
12	Rheumatoid arthritis plus premature family history, 10-year ASCVD <3%, 30-year ASCVD unavailable or <15%, without established diabetes, a kidney signal or severe hypercholesterolemia.	3A LEVEL_3A_INFLAMMATORY_CONTEXT
13	At least two enhancer-burden flags plus at least one metabolic-burden flag.	3B LEVEL_3B_MULTIPLE_ENHANCERS
14	Elevated Lp(a) plus at least two metabolic-burden flags.	3B LEVEL_3B_LPA_METABOLIC_BURDEN
15	Current smoking: established diabetes or at least two actionable drivers gives 3B; otherwise 3A.	3B or 3A LEVEL_3B_SMOKING_PLUS_BURDEN LEVEL_3A_SMOKING
16	BORDERLINE, INTERMEDIATE or HIGH 10-year PREVENT ASCVD category, plus at least two actionable drivers, albuminuria, or at least three mild signals.	3B LEVEL_3B_PREVENT_PLUS_BURDEN
17	HIGH PREVENT category, established diabetes, or a clinical kidney pathway (established CKD or current G3a-G5). Compatibility path uses a qualifying kidney signal.	3B LEVEL_3B_ACTIONABLE_CLINICAL_PATHWAY
18	Kidney confirmation pathway: current A2/A3 or documented albuminuria without an earlier clinical kidney pathway.	2A LEVEL_2A_ISOLATED_KIDNEY_CONFIRMATION
19	ApoB ≥120 mg/dL.	3A LEVEL_3A_APOB_GE_120
20	At least two counted mild signals.	2B LEVEL_2B_MULTIPLE_MILD_SIGNALS
21	Exactly one counted mild signal.	2A LEVEL_2A_SINGLE_MILD_SIGNAL
22	PREVENT ASCVD unavailable and one or more listed blocking inputs missing: systolic BP, total cholesterol, HDL-C, eGFR or BMI.	Incomplete LEVEL_INCOMPLETE_MISSING_PREVENT_INPUTS

Ordered base rules, engine reference 18baf7a

Order	Condition	Base output and rule identifier
23	No earlier rule matches.	1 LEVEL_1_MINIMAL_RISK

The 3C modifier

After any base 3B rule, the engine checks 10-year PREVENT ASCVD ≥5%, established diabetes or unconfirmed diabetes-range A1c, and the kidney-signal helper. Any positive condition changes the displayed Level to 3C, retains the base rule identifier, and records the convergence drivers separately. RSS is not an input to this modifier.

Definitions used by the counters

• **Early glycemic risk:** the canonical glycemia interpretation recognizes prediabetes or unconfirmed diagnostic-range A1c. Established diabetes is a separate state. A risk signal alone does not establish a diagnosis.

• **CKM context:** BMI ≥30 kg/m², OSA or MASLD, counted together as one context flag.

• **BP signal:** represented BP treatment, hypertension/elevated-BP state, or at least two usable BP pairs with systolic ≥130 or diastolic ≥80 mmHg. The Level helper's BP counter is distinct from RSS's disease-authority requirement.

• **Elevated Lp(a):** ≥125 nmol/L or ≥50 mg/dL in the reported unit. No fixed conversion between units is applied.

• **Kidney signal:** established CKD, current G3a-G5, current A2/A3, or supported problem-list albuminuria. The compatibility path uses eGFR <60, UACR ≥30, or documented albuminuria. Current abnormal categories can trigger a signal while chronicity remains unconfirmed.

• **Counted mild signals:** early glycemic risk; TG ≥150; ApoB 80-<120, or, when ApoB is absent, atherogenic findings (valid LDL-C ≥100, non-HDL-C ≥130 or TG ≥150); CKM context; elevated Lp(a); premature family history; reproductive history; inflammatory disease; HIV; South Asian ancestry; Filipino ancestry; breast arterial calcification; hsCRP ≥2; and BORDERLINE or INTERMEDIATE PREVENT ASCVD category. The lipid fallback is one flag and can overlap with the separate TG flag.

• **Actionable drivers:** albuminuria, reduced current G category, established diabetes, smoking, LDL-C 160-189.999, severe hypercholesterolemia, suspected FH, BP signal, ApoB ≥100 (one ApoB driver, with a distinct label at ≥120), elevated Lp(a), premature family history, reproductive history, early glycemic risk, TG ≥150 and CKM context.

• **Metabolic-burden counter:** one flag each for early glycemic risk, established diabetes, BP signal, CKM context, TG ≥150, ApoB ≥100, non-HDL-C ≥160 and valid LDL-C ≥130. These flags can describe

related biology.

• **Enhancer-burden counter:** elevated Lp(a), premature family history, any reproductive flag, inflammatory disease, HIV, active cancer, cancer survivorship with represented life expectancy >2 years, South Asian ancestry, Filipino ancestry and breast arterial calcification. Each is a separate Boolean flag in this implementation.

• **Hidden-risk enhancer counter:** elevated Lp(a), premature family history, inflammatory disease, South Asian ancestry, Filipino ancestry, gestational diabetes, preeclampsia, early/premature menopause (one flag) and breast arterial calcification.

• **Inflammatory signal:** a generic inflammatory-disease flag, rheumatoid arthritis, SLE, psoriasis, inflammatory arthritis or IBD. **Reproductive flags:** early/premature menopause, preeclampsia, gestational hypertension/diabetes, preterm delivery, small-for-gestational-age infant, recurrent pregnancy loss, PCOS/irregular menses and early menarche.

10-year PREVENT ASCVD categories are Low <3%, Borderline 3-<5%, Intermediate 5-<10%, and High ≥10%. In the released reference revision, the classifier retains governed unrounded precision at the 3% and 5% boundaries when matching raw model provenance is available; displayed probabilities remain rounded. The 10% boundary retains its existing reported-value behavior in that revision.

Separate numerical correction under review. Application draft PR #80 extends the same provenance-checked unrounded policy to 10%. Calculated 9.996% would remain Intermediate even when the reported numeric value is 10.00%; exactly 10% and clinician-provided 10% without trustworthy calculated provenance remain High. The equations and reported numeric values do not change. This correction can change downstream Level or action decisions and requires review of the exact clinical differences and release checks before deployment.

The separate presentation draft uses one-decimal probabilities and a category-preserving label such as “Just below 10%” when ordinary rounding would obscure the boundary. For example, 9.96% is already Intermediate under the released rules, but ordinary one-decimal display would show 10.0%. That display issue is distinct from the 9.996% classification correction. No additional near-threshold clinical category is introduced.

Physician-agreed Level refinement, not released. A separate revision to the numerical proposal sets a minimum of Level 3A for Intermediate 10-year PREVENT ASCVD risk (5-<10%). The existing base rules run first; only a Level 1, 2A or 2B with an authoritative Intermediate category is raised by LEVEL_3A_INTERMEDIATE_PREVENT. Existing 3A pathways and stronger driver, plaque or clinical-disease assignments remain in place. The 3A definition becomes “Modeled risk or single driver,” with the specific rule and explanation identifying the pathway. This supersedes the earlier plan to preserve every legacy Level assignment. It is an RCCKM taxonomy choice, not an AHA-defined Level or a calibrated event-risk ranking. PREVENT probability/category, CKM stage, KDIGO, RSS and treatment decisions remain independently determined.

The synthetic calculated 9.996% example therefore changes from released High/3C to proposed Intermediate/3A, rather than Intermediate/2B under the precision correction alone. The calculated probability is unchanged; no buffer is added around 10%. Exactly 10% remains High. This Level minimum does not itself change a statin-intensity rule, establish kidney chronicity or supply missing PREVENT

inputs. The physician owner approved the exact clinical differences and revised baseline on November 4, 2026. The refinement is committed to draft PR #80 and awaits CI and release validation.

All thresholds above document implementation choices for review. The Level framework and its rule counters have not been fitted to an outcome dataset or independently validated as a new risk-prediction model.

A visible tally of selected findings.

RSS v2 is a prespecified ordinal burden score. Points reflect an intended hierarchy: plaque receives greater weight, major biologic states receive intermediate weight, and contextual findings receive smaller weights. Exact weights are RCCKM design choices informed by clinical evidence; they are not fitted regression coefficients, converted hazard ratios or probabilities.

The displayed total is the sum of displayed positive contributions. Findings can remain visible as context when a cap limits their points. Missing measurements contribute no points and remain unknown; a low total does not establish that unmeasured domains are normal.

How overlap is handled

• **Lipids:** ApoB, or LDL-C ≥190 when ApoB is absent, is allocated first; suspected FH next; triglycerides last. These contributions share a 12-point cap. Lp(a) is a separate measured domain.

• **Plaque:** a numeric CAC result supersedes qualitative incidental CAC. The two do not stack.

• **BP:** disease authority or repeated credible elevations is required. Diagnosis, therapy and current measurement produce one contribution, up to 8 points. An isolated reading alone does not qualify.

• **Inflammatory disease and ancestry:** multiple supported labels within each domain contribute once. HIV remains a separate domain; hsCRP has its own measured contribution.

• **Reproductive history:** 2 points per supported marker, up to 4. Early and premature menopause are reconciled into one menopause item.

• **Kidney findings:** eGFR and albuminuria are additive, reflecting separate G and A axes. Their contributions do not by themselves establish CKD chronicity.

Technical appendix B · Every RSS band and point value

These tables are generated from the reviewed engine's canonical RSS methodology export. The downloadable JSON also supplies exact continuous interval endpoints and source fingerprints. Integer-style band labels below follow the engine's display convention.

CAC

CAC: prespecified RSS weights

Finding	Points	Scoring note
1-99	10	CAC 1-99 indicates mild plaque burden.
100-299	20	CAC 100-299 indicates moderate plaque burden.
300-999	30	CAC 300-999 indicates severe plaque burden.
>=1000	40	CAC >=1000 indicates extreme plaque burden.

ApoB

ApoB: prespecified RSS weights

Finding	Points	Scoring note
80-99 mg/dL	4	ApoB 80-99 represents borderline elevation.
100-129 mg/dL	8	ApoB 100-129 represents elevated ApoB.
>=130 mg/dL	12	ApoB >=130 is a higher RCCKM RSS particle-burden scoring band, not a treatment threshold.

LDL-C fallback

LDL-C fallback: prespecified RSS weights

Finding	Points	Scoring note
LDL-C >=190 mg/dL when ApoB unavailable	10	LDL-C >=190 mg/dL indicates severe hypercholesterolemia.

Lp(a)

Lp(a): prespecified RSS weights

Finding	Points	Scoring note
75-124 nmol/L / 30-49 mg/dL	2	Lp(a) 75-124 nmol/L is the consensus grey zone and a small risk-refinement signal.
125-249 nmol/L / 50-99 mg/dL	8	Lp(a) 125-249 nmol/L indicates elevated Lp(a).
250-429 nmol/L / 100-179 mg/dL	12	Lp(a) 250-429 nmol/L indicates higher elevated Lp(a).
>=430 nmol/L / >=180 mg/dL	15	Lp(a) >=430 nmol/L indicates very high Lp(a) elevation.

hsCRP / inflammatory risk

hsCRP / inflammatory risk: prespecified RSS weights

Finding	Points	Scoring note
Single hsCRP >=2 mg/L without established persistence/context	2	Unconfirmed measurement; confirm persistence or clinical context.
2-4.9 mg/L	4	hsCRP 2-4.9 mg/L indicates elevated inflammatory risk.
>=5 mg/L	7	hsCRP >=5 mg/L indicates high inflammatory risk.

Blood pressure / hypertension

Blood pressure / hypertension: prespecified RSS weights

Finding	Points	Scoring note
Established/treated and controlled or current BP unavailable	3	Underlying hypertension burden persists despite control; an isolated reading alone does not qualify.
Established/repeated stage 1 BP	4	Mild established BP burden.
Established/repeated stage 2 BP	6	Major established BP burden.
Repeated markedly/severely elevated BP	8	Maximum BP-domain burden; acute action remains separate from RSS.

Kidney disease

Kidney disease: prespecified RSS weights

Finding	Points	Scoring note
45-<60	5	eGFR 45-<60 indicates mild kidney function reduction.
30-<45	8	eGFR 30-<45 indicates moderate kidney function reduction.
15-<30	12	eGFR 15-<30 indicates severe kidney function reduction.
<15	15	eGFR <15 indicates kidney failure risk.
30-<300 mg/g	6	UACR 30-<300 indicates moderately increased albuminuria.
>=300 mg/g	10	UACR >=300 indicates severely increased albuminuria.

Triglycerides

Triglycerides: prespecified RSS weights

Finding	Points	Scoring note
150-499 mg/dL	4	Triglycerides 150-499 indicates hypertriglyceridemia.
>=500 mg/dL	4	Triglycerides >=500 indicates severe hypertriglyceridemia; pancreatitis urgency is handled outside RSS.

Diabetes / dysglycemia

Diabetes / dysglycemia: prespecified RSS weights

Finding	Points	Scoring note
5.7-6.4%	2	A1c 5.7-6.4% indicates prediabetes-range dysglycemia.
Diabetes / diabetes-range glycemia	8	Diabetes is a risk enhancer for ASCVD.

Smoking and family history

Smoking and family history: prespecified RSS weights

Finding	Points	Scoring note
Current smoking	8	Current smoking is a modifiable risk factor.
Premature family history	3	Premature first-degree family history is a risk enhancer.
Suspected FH / HeFH	8	Up to 8 points, limited by the 12-point atherogenic-lipid overlap cap.

Incidental CAC

Incidental CAC: prespecified RSS weights

Finding	Points	Scoring note
Present	10	Incidental coronary calcium is qualitative plaque evidence.
Severe	20	Severe incidental coronary calcium is qualitative high-burden plaque evidence.

Contextual enhancers

Contextual enhancers: prespecified RSS weights

Finding	Points	Scoring note
Chronic inflammatory disease, HIV, OSA, MASLD, or higher-risk ancestry	2	Each distinct contextual domain is scored once.
Reproductive risk markers	2 each; maximum 4	The reproductive-history domain is capped so correlated markers do not stack without limit.

Interpretation details

For BP, the implementation classifies stage 1 at systolic ≥130 or diastolic ≥80, stage 2 at ≥140 or ≥90, markedly elevated at ≥160 or ≥100, and severely elevated at ≥180 or ≥120 mmHg, with higher categories evaluated first. One severe current reading in established hypertension receives 6 points; 8 requires at least two markedly/severely elevated historical readings. Below-stage-1 or unavailable current BP with established disease receives 3. Clinical urgency is evaluated outside RSS.

A single hsCRP ≥2 mg/L without persistence or supportive chronic context receives 2 points. The 4- and 7-point bands require the corresponding persistence/context authority. Established diabetes or A1c ≥6.5% receives the 8-point glycemic contribution; an isolated diagnostic-range result remains subject to separate diagnosis-confirmation rules.

RSS's contextual ancestry domain represents supported South Asian and/or Filipino flags once. RSS does not use race or ancestry as a PREVENT coefficient. Contextual weighting, including ancestry, OSA and MASLD, requires further evaluation for calibration, fairness and incremental usefulness.

Reproductive items include supported menopause and menarche age/flags, preeclampsia, gestational hypertension/diabetes, preterm delivery, small-for-gestational-age infant, recurrent pregnancy loss and PCOS/irregular menses. A reproductive item adds 2 points only within the 4-point cap; all represented items remain available as detail.

RSS has not undergone external prognostic validation or formal recalibration. Its point values should be reviewed as a transparent visualization convention alongside the individual findings, PREVENT and established disease context.

Each scenario states how it was modeled.

Each plan forks the reviewed worksheet and reruns the clinical engine. The Level, RSS and domain interpretation reflect that scenario. The ASCVD probability comparison uses two explicitly distinguished methods.

Trial-derived LDL-C and systolic BP scenarios

For LDL-C lowering, the simulator uses the CTT 2010 relative-effect estimate of 0.78 per 1 mmol/L (38.67 mg/dL) reduction. For systolic BP lowering, it uses Ettehad 2016's estimate of 0.80 per 10 mmHg reduction. [4] [5] The engine scales each factor to the selected change, multiplies the factors, and applies them to baseline survival:

```
f_lipid = 0.78 ^ ((LDL_before - LDL_plan) / 38.67)

f_BP = 0.80 ^ ((SBP_before - SBP_plan) / 10)

f_combined = product of applicable factors

p_plan = 1 - (1 - p_reference) ^ f_combined
```

Here, `p_reference` is a proportion between 0 and 1. With lipid/BP changes alone, it is the baseline PREVENT ASCVD estimate. In a mixed scenario, supported nontrial changes first recalculate PREVENT, then the lipid/BP factors are applied. Current code rounds each factor to three decimals before multiplication and reports risk percentages to two decimal places. Those decimal places describe numerical output precision; they do not quantify the uncertainty of the projection. Changes smaller than 0.5 mg/dL LDL-C or 0.5 mmHg systolic BP do not add a trial-effect factor.

The presentation draft reduces visible probabilities to one decimal while preserving underlying numeric results and audit provenance. It also removes improvement/worsening comparisons based on Level ordering. No uncertainty interval is displayed without an established method for deriving one.

The synthesis adds modeling assumptions.CTT studied major vascular events, including revascularization; Ettehad studied major cardiovascular outcomes, including heart failure. These endpoints and populations differ from PREVENT ASCVD. RCCKM treats the published relative effects as proportional hazard factors, assumes multiplicative combined effects, and applies them to the selected PREVENT horizon. This combined projection has not been directly validated in a clinical outcomes study.

Recalculated risk-factor scenarios

Supported smoking, HbA1c and UACR changes rerun the applicable PREVENT model. A new HbA1c or UACR value can affect probability only if that predictor belongs to the baseline model. The comparison preserves the baseline model and diabetes status; a change that would alter either is withheld with an explanation. These comparisons describe a modeled risk profile and do not represent the cardiovascular or kidney benefit of a particular drug.

Prespecified treatment assumptions

Starting a moderate-intensity statin assumes LDL-C and non-HDL-C decrease by 35%, and ApoB by 30%, with HDL-C held constant for the non-HDL calculation. Intensification starts from the **current on-treatment lipid value**. For a moderate-intensity regimen, the LDL/non-HDL multiplier is 0.50/0.65, approximately 0.769: the assumed remaining fractions of an untreated value on high- and moderate-intensity therapy. A current LDL-C of 100 mg/dL therefore becomes approximately 77 mg/dL, an additional reduction of about 23%. ApoB is multiplied by 0.80. For a represented low-intensity regimen, the current LDL/non-HDL value is multiplied by 0.50/0.80 (0.625), and current ApoB by 0.70. Lipid values are rounded as implemented. These are scenario defaults; observed response requires follow-up.

If LDL-C is absent, the selected total-cholesterol change, with baseline HDL-C held constant, can serve as a labeled non-HDL-C proxy for the lipid effect. ApoB affects interpretation but receives no additional trial ASCVD reduction. Treatment effects and direct predictor changes are separated to avoid counting the same lipid/BP change twice.

Horizon and scope

RCCKM reports supported 10-year PREVENT estimates at ages 30-79 and 30-year estimates at ages 30-59. Its current clinical eligibility policy withholds PREVENT for established clinical cardiovascular disease, established heart failure and CAC ≥300; clinicians still review the source guidance and exclusions beyond those represented in software. [1]

The scenario comparison models ASCVD. It does not project total CVD, heart failure, annual event timing, life expectancy, statin withdrawal or a complete benefit-harm balance. Carrying a trial-derived effect to 30 years assumes sustained benefit beyond the roughly five-year statin trial follow-up. Connecting lines between 10- and 30-year points are visual guides. Adherence, changing health, competing risks and

treatment changes can alter an individual's course.

A shared engine, with clinician review in the loop.

1. **1 · Reconcile the source**A rule-based parser proposes structured values, dates, histories and medication states from supported chart text. Missing fields and conflicts remain visible.

2. **2 · Confirm the worksheet**The clinician reviews or edits the proposed inputs. Structured disease, measurement, treatment and certainty states carry their own meanings.

3. **3 · Run the clinical engine**The Python engine evaluates inputs, published calculations and explicit domain rules, producing structured results and traceable reasons.

4. **4 · Review and communicate**The interface presents the assessment, scenarios, clinician note and patient handout. The clinician reviews these outputs and makes the clinical decisions.

Deterministic core and optional AI

Confirmed inputs, the same engine/evidence version and the same evaluation context follow the same deterministic calculation and decision logic. Next.js, FastAPI and the retained Streamlit interface consume the shared engine's results. Renderers select and format structured clinical facts rather than independently recomputing them.

The optional Clinical Pearl is separately labeled AI commentary. It is generated from assessment context for clinician review and can vary. It does not supply the deterministic PREVENT calculation, Level, RSS or staging. Human approval remains part of using the note, diagnosis support and patient handout.

Input meaning and assessment lifecycle

The engine distinguishes absent results from measured zero, quantitative results from bounded/nonnumeric results, current treatment from documented non-use, and established disease from a finding that needs confirmation. Clinical context and observation dates matter. Input changes invalidate a stale assessment and dependent outputs, prompting reassessment.

Kidney interpretation retains G/A measurement categories separately from CKD certainty. Glycemia separates established diabetes from unconfirmed diagnostic-range measurements. BP interpretation separates a usable reading, disease authority and treatment context. These distinctions let an abnormal result prompt review without silently creating a confirmed diagnosis.

Versioning and interoperability

Result metadata includes application version, build identifier, rule-set version, evidence-map hash and generation timestamp. Rule traces expose triggering inputs and rule identifiers where implemented. A reproducible review must specify the software/evidence version, confirmed inputs, evaluation date and relevant configuration.

The current public workflow uses synthetic or de-identified text/manual entry and clinician-controlled output transfer. FHIR R4 normalization, SMART sandbox and CDS Hooks implementation are separate integration capabilities. Their presence does not establish a live production EHR connection or institutional authorization.

Evidence is recorded with its scope.

Automated testing evaluates calculations, defined rules and workflow behavior against specified expectations. Synthetic cases make boundaries and regressions reproducible. These checks do not establish clinical effectiveness, predictive calibration of the proprietary frameworks, or benefit in actual patient care.

Developer-maintained evidence register, checked November 4, 2026; dates are Eastern Time

Record	Observed result	Scope and limits
Routine validation November 3, 2026 18baf7a	Full pytest suite on Ubuntu: **7,240 passed, 160 skipped, 2 warnings**. The overall multi-job workflow completed successfully.	Includes parser, clinical-rule, boundary, invariant, snapshot and related checks. Separate workflow jobs cover Next.js build/type checks and browser regressions. Skipped tests are not counted as passes.
Provider-free deterministic replay November 3, 2026 18baf7a	**PASS**: 500 primary synthetic patients plus 100 format variants; reference, approved-baseline comparison and frozen-source identity checks completed.	600 journeys across 500 distinct synthetic patients. This replay checks defined contracts against the governed reference/baseline; it does not run a new live AI clinical review.
Browser + semantic QA October 31, 2026 933e1f6	**PASS**: 500 primary and 100 paired journeys completed; 600 AI reviews completed; zero deterministic failures, no recorded severity findings and no blockers.	Recorded for a frozen earlier source through the retained Streamlit driver. AI semantic QA is distinct from independent human expert validation. This entry does not assert a full Next.js clinical semantic PASS or a new semantic PASS for 18baf7a.

These figures were checked against the repository's workflow records and job logs. The downloadable evidence register retains full revision identifiers, run identifiers, receipt excerpts and scope notes. Source code and full CI artifacts are access-controlled; a reviewer can request supporting evidence. The register

is a developer-maintained record, not independent certification.

What the verification layers examine

• **Calculation checks:** PREVENT numerical comparisons against an independently executed, governed reference implementation and supported model/eligibility contracts.

• **Clinical-rule and boundary checks:** synthetic archetypes and threshold cases for disease, plaque, lipids, glycemia, kidney, BP, missing information and output wording.

• **Regression and invariant checks:** changes to parsed inputs, structured results and copied reports against approved expectations. A changed expectation requires review; matching a baseline alone does not prove the baseline is clinically correct.

• **Browser journeys:** ingestion, reconciliation, worksheet state, assessment presentation and copied output. The corpus includes 100 meaning-preserving format variants in addition to 500 primary cases.

• **AI semantic QA:** a separately executed model reviews synthetic journey evidence using a specified rubric. Findings and infrastructure failures are retained. Model-based review has its own limitations and does not substitute for human clinical adjudication.

Clinical questions still to be answered

Evaluation priorities; these are questions, not reported study results

Question	Approach to evaluate it
Are interpretations clinically appropriate?	Prespecified case review by independent clinicians, with disagreement, severity and adjudication reported separately from software pass rates.
Does the workflow help clinicians?	Usability, task completion, decision time, workload and comparison with usual workflow. Any claim of time saved requires direct measurement.
Will clinicians continue using it?	Prospective usage and retention, with enrolled-clinician denominators, a prespecified sustained-use definition and incomplete use retained in the analysis.
Does the conversation help patients?	Patient understanding and shared decision-making measures, with response rates and potential clinician-administered response bias considered.
Do Levels or RSS add prognostic value?	Independent outcome-linked datasets, calibration/discrimination and incremental-value analyses appropriate to any proposed predictive claim.

Independent prospective evidence that RCCKM improves clinical decisions, workflow efficiency, shared decision making or patient outcomes has not yet been established. PREVENT's published validation supports its source equations; it does not validate RCCKM's additional interpretation layers or the combined simulator projection.

Technical appendix C · Version binding and evidence governance

Development checks, merge checks and clinical-release checks have different scopes. Routine CI includes deterministic replay, baseline comparison and source-identity checks. Full browser/AI review is a separate explicitly invoked workflow. The presence of workflow files alone does not establish live branch protection or deployment enforcement; those settings are not certified by this document.

Full-run evidence is bound to an immutable candidate, corpus, numerical reference, contract, rubric, prompt, model and runner identity. Results from different candidates or interfaces must not be combined into a new PASS. Earlier adverse findings and their adjudications remain part of the development record, separate from later passing runs.

The semantic record above uses source SHA-256 `ae6c5c7fffdfb6a2d4741ba57a5e68233a9c7a20ae884e3d7a584eebd1bd6af6`. It covers revision `933e1f6e6c1dbbbd5fd1af5ed84566c6ea3cbb45`. The current methods reference is `18baf7a8651b10cb4f047072590a5c8699e293c5`. Its routine/deterministic evidence is reported separately. These identifiers document reviewed source; they do not independently establish the version currently served by a hosted application.

Use the output within the full clinical picture.

RCCKM is intended to support clinician review and prevention discussions. Correct interpretation depends on the accuracy, completeness, dates and provenance of the inputs. A parser can miss or misread information, a clinical rule can omit a relevant circumstance, and synthetic testing cannot cover every patient or chart format.

Levels, RSS and simulator assumptions have their own evaluation needs. Some rules count correlated findings, and fixed weights cannot establish individualized treatment benefit. A Level name does not supply a diagnosis, a calibrated event probability or a complete treatment decision. Contraindications, adverse effects, tolerability, cost, treatment burden and patient preferences require clinician-patient discussion.

Public demonstration uses synthetic or properly de-identified information. Do not enter patient identifiers. Optional AI features have separate data-processing implications; consult the current Privacy Policy and Terms. Identifiable-data use and production institutional integrations require their own infrastructure, agreements and approvals.

Published foundations. Explicit implementation choices.

1. American Heart Association. PREVENT equations and clinical resources.
Published equation framework, primary-prevention use and optional predictors. Includes the current clinician FAQ and source-method links.

2. American Heart Association. Cardiovascular-Kidney-Metabolic Health Presidential Advisory. 2023.
Integrated CKM framework and staging.

3. KDIGO. 2024 Clinical Practice Guideline for the Evaluation and Management of CKD.
Kidney classification and chronicity. This methods version refers to the 2024 guideline; a public-review draft does not replace the implemented source.

4. CTT Collaboration. Efficacy and safety of more intensive lowering of LDL cholesterol. Lancet. 2010;376:1670-1681.
Trial-derived LDL-C effect used in scenario modeling. DOI: 10.1016/S0140-6736(10)61350-5.

5. Ettehad D, et al. Blood pressure lowering for prevention of cardiovascular disease and death. Lancet. 2016;387:957-967.
Trial-derived systolic BP effect used in scenario modeling. DOI: 10.1016/S0140-6736(15)01225-8.

6. 2026 ACC/AHA multisociety Guideline on the Management of Dyslipidemia.
Lipid, biomarker and CAC prevention context. Guideline support for a finding does not validate the RSS weight assigned to it.

7. European Atherosclerosis Society. Lipoprotein(a) consensus statement. 2022.
Unit-specific Lp(a) concentration context; RSS weights remain prespecified platform choices.

8. Budoff MJ, et al. Ten-year association of coronary artery calcium with ASCVD events: MESA. 2018.
Observed CAC risk gradients; the RSS point scale itself is not derived or validated by this study.

References identify underlying scientific work. RCCKM is independently developed by Risk Continuum Health LLC. References do not imply organizational endorsement.

Version 1.0 · November 4, 2026. Initial public methods document: Level hierarchy and counters, engine-exported RSS v2 weights, simulator transformations, engineering boundaries and dated evaluation register. Engine source reference: `18baf7a8651b10cb4f047072590a5c8699e293c5`. The downloadable JSON records SHA-256 fingerprints of the reviewed implementation files. Clinical rule changes require a methods review and an updated document version.

Documentation scope: an implementation description and evaluation summary. Complete source code, every domain's treatment algorithm and an independent certification are outside this document's scope.